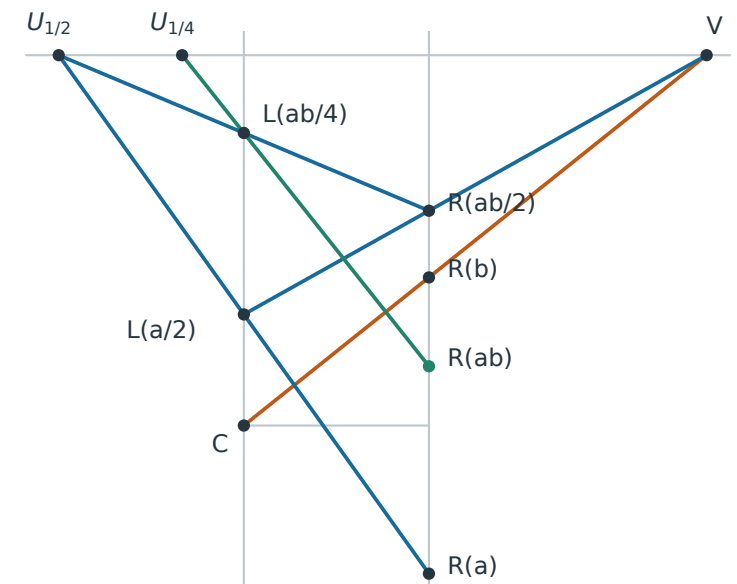
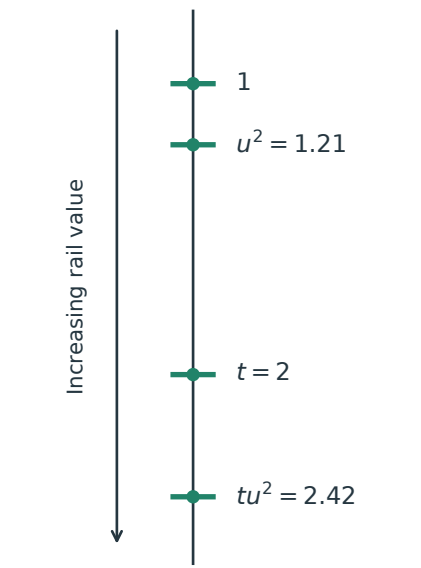


Ruler-only product: five joins



Example: $a = 1.4$, $b = 0.6$, $ab = 0.84$
If $b = 1$, reuse $R(a)$: zero joins.

Gate on one oriented rail



$1 \leq t$, $u^2 \leq t$, $1 \leq tu^2$
Three comparisons: accept this trial.